

● WEEK 1 · SESSION 1 · COMPANION E-BOOK

UNDERSTAND AI.

*See the Landscape Clearly — the week
you stop guessing about AI and start
seeing how it actually works.*

SESSION 1

How AI actually works — under the hood.

FOR EVERYONE

Written for beginners and professionals at once.

YOU LEAVE WITH

A working model of how AI functions.

WELCOME TO WEEK 1

WEEKLY OBJECTIVE

You had opinions about AI. By the end of this week you have a working model of how it actually functions — technically, economically, and professionally.

Identity shift — AI Curious → AI Aware

THIS BOOK · SESSION 1

SESSION 1

HOW AI ACTUALLY WORKS

From weights and next-token prediction to training, cost, and why Claude behaves the way it does.

HOW EVERY TOPIC IS BUILT

This is your Week 1 companion — read it alongside the two live sessions. It is written for two people at once: the total beginner who has only ever typed into ChatGPT, and the professional who wants the real mechanics. Skim the plain-English parts; stop on the prompts and the build steps.

WHAT	A plain-English definition of the idea.
WHY	Why it matters to you, at work and in the wallet.
WHERE	Where you actually meet it in day-to-day tools.
WHEN	When to think about it — and when to ignore it.
HOW	How it works under the hood, still in plain words.

Every topic also carries: advantages and disadvantages, a real example, three professional prompts at three levels, step-by-step build-up and execution, and the odd Fun Fact.

WHAT'S IN SESSION 1

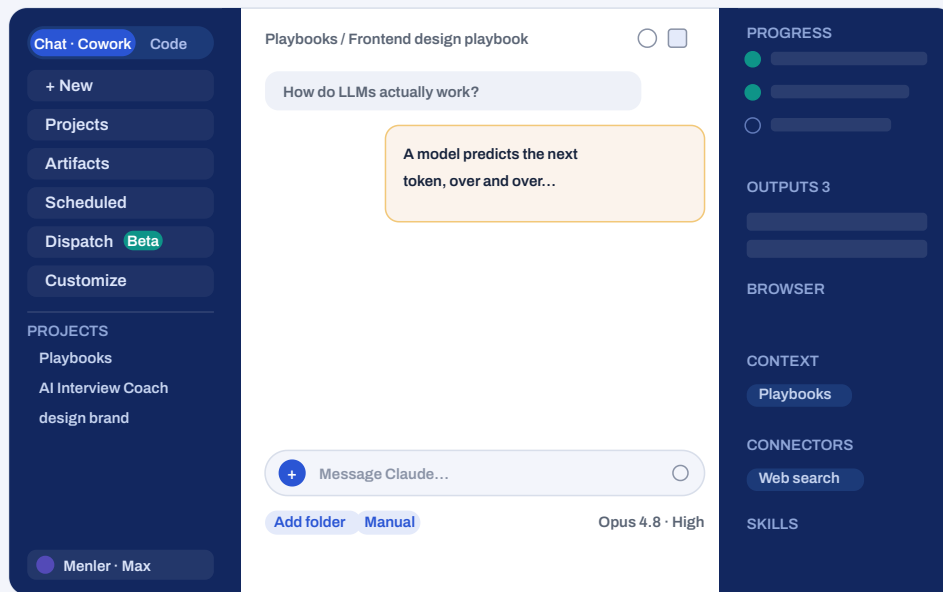
A full tour of the Claude app (3 pages), then the four Session 1 topics — how AI actually works under the hood. Jump to whatever you need.

◆	Getting around Claude	p. 4
	The whole interface + how to connect & automate	
1.1	How LLMs Really Work	p. 7
	A language model is a very, very good guesser of the next word.	
1.2	The Language of AI	p. 10
	Tokens, context windows, temperature, embeddings — the four words that unlock the rest.	
1.3	How AI Learns	p. 13
	Pre-training, fine-tuning, and RLHF — the three stages that turn raw text into a helpful assistant.	
1.4	Reading the Scoreboard	p. 16
	Benchmarks, the 3-way comparison, and why Claude leads for generalists.	

BEFORE YOU START · THE CLAUDE-FIRST TOUR

GETTING AROUND CLAUDE

This whole program runs on Claude. Learn the app once — here is the entire window, every part labelled.



THE CLAUDE APP — LEFT SIDEBAR, TOP BAR, CHAT (CENTRE), MESSAGE BOX (BOTTOM), AND THE LIVE PANEL (RIGHT)

THE LEFT SIDEBAR — WHERE YOU MOVE BETWEEN KINDS OF WORK

New — Start a fresh chat or Cowork session. Nothing carries over unless the session shares a Project.

Artifacts — Your gallery of built pages and docs (dashboards, mini-apps, reports) that live on beyond the chat and can be re-opened or shared.

Dispatch — Hand a job to a background agent that plans and runs it while you do other things. (Beta.)

Projects list — Your projects pinned for quick access; click one to jump straight into its space.

Projects — Your Projects — shared folders of chats, files and instructions that persist across sessions and your team.

Scheduled — Tasks that run on a schedule — hourly, daily, weekly. View, pause, edit or run them here.

Customize — Your home for connectors, custom instructions, styles and preferences — set once, applies everywhere.

Account · plan — Bottom-left: who you are and your plan (e.g. Max). The download icon gets the desktop app.

THE TOP BAR — WHERE YOU ARE, AND HOW TO GET AROUND

Chat · Cowork / Code — The mode switch. “Chat & Cowork” for everyday work with files and tools; “Code” opens Claude Code, the developer surface.

Back / forward — Move through screens you have visited, like a browser.

Panel & theme — Top-right icons: show or hide the side panels and switch light / dark.

Search — Find any past chat, project or output by keyword.

Breadcrumb — Shows where you are (e.g. Playbooks / Frontend design...). Click a crumb to go up a level.

THE MESSAGE BOX — HOW YOU BRIEF CLAUDE AND CONTROL IT

The “+” — Add connectors, upload files, and switch on skills for this message.

Approval: Manual / Auto / Skip — When Claude asks before acting. Manual = approve every action; Auto = auto-approves reads, checks writes; Skip = no prompts (trusted work only).

Model · reasoning — Pick the model (e.g. Opus 4.8) and the thinking effort — High for hard problems, lower for speed.

Add folder — Give the session a folder from your computer to read and write — pick the smallest folder that holds what is needed.

Dictate — The mic — talk instead of type.

THE PANEL ON THE RIGHT — WHAT CLAUDE SHOWS YOU WHILE IT WORKS

Progress — The live task list — every step Claude is taking, ticking off as it goes.

Browser · Claude in Chrome — The browser Claude is driving for web tasks, when a browser is connected.

Connectors — The apps switched on for this session (e.g. Web search); add more from “+” or Customize.

Outputs — The finished files from this session — open, preview, download, or send some to Drive.

Context — What this session can see — the connected Project and folders (e.g. Playbooks).

Skills — The packaged know-how Claude can apply here on demand.

◆ FIND IT IN CLAUDE

Message box → Manual / Auto / Skip

Set the approval mode before a big task. New to Cowork? Start on **Manual** so you approve each action. As you trust a routine, move to **Auto**. Keep **Skip** for small, contained jobs only.

THE THREE THINGS THAT TURN CLAUDE FROM A CHATBOT INTO A WORKSPACE

Anyone can type into Claude. The step up is **connecting your own apps**, **putting work on repeat**, and **handing off jobs to the background**. Here is where each lives and how to switch it on.

1

CONNECT YOUR APPS

CONNECTORS

A connector plugs one of your apps — Gmail, Google Drive, Notion, Slack, and hundreds more — straight into Claude, so it can read and act on that data instead of you copy-pasting.

- 1 Open Settings → Connectors (or, in a chat, click “+” → Connectors → Manage connectors).
- 2 Click the “+” next to Connectors to open the directory, then pick the app you want.
- 3 Click Connect, then sign in and approve access in the pop-up (this is OAuth — you approve, Claude never sees your password).
- 4 Back in any chat, click “+” → Connectors and toggle the app on for that conversation.

Free, Pro, Max, Team & Enterprise. Team/Enterprise: an Owner enables a connector for the org first. Choose the least access that does the job.

2

PUT WORK ON REPEAT

SCHEDULED TASKS

A scheduled task is a prompt Claude runs on its own on a set cadence — a Monday digest, a daily inbox triage — and reports back with no nudge from you.

- 1 In the left sidebar, click Scheduled.
- 2 Click New task → Create with Claude (guided) or Set up manually.
- 3 Write the prompt as a full standalone instruction, pick the cadence (hourly / daily / weekly / weekdays), and set the approval mode + folder.
- 4 Click Schedule. Pause, edit, or run it on demand any time from the same page.

Pro, Max, Team & Enterprise. Desktop for all paid plans; web & mobile in beta.

3

HAND OFF, WALK AWAY

DISPATCH

Dispatch is a background agent. You describe an outcome once; it breaks the job into tasks, runs each on its own, and surfaces the results later — even from your phone.

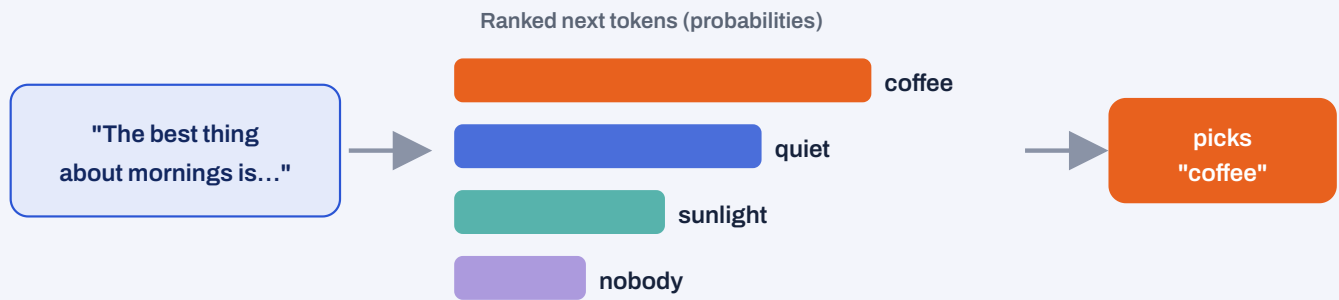
- 1 In the left sidebar, select Dispatch.
- 2 Describe the outcome like you would brief a colleague, in one message.
- 3 Let it plan — it starts child tasks that each show a status (Running, Awaiting input, Completed...).
- 4 Check back later: open any task to read the transcript and grab the files it produced.

Pro & Max, on the latest Claude Desktop (macOS/Windows). Keep the desktop app open to assign from mobile.

1.1 SESSION 1

HOW LLMS REALLY WORK

A language model is a very, very good guesser of the next word.

**WHAT**

A Large Language Model (LLM) like Claude is a program that has read a huge slice of human writing and learned one skill extremely well: given some text, predict the next small chunk of text. That chunk is called a **token**. It does this over and over — predict a token, add it, predict the next — and out comes a sentence, a paragraph, an essay.

WHY

Once you understand that AI is **predicting**, not **looking up**, everything else makes sense: why it can be confidently wrong, why how you ask changes what you get, and why it is a brilliant thinking partner but a shaky encyclopaedia. This one idea removes most of the confusion beginners carry.

WHERE

Every chat tool you touch — Claude, ChatGPT, Gemini — is an LLM doing next-token prediction. The chat box is just a friendly wrapper around this guessing engine.

WHEN

Think about it whenever the model surprises you. A weird answer is usually not a “bug” — it is the most probable next words given how you phrased things. Change the phrasing, change the probability.

HOW IT WORKS

During training the model saw billions of examples and tuned millions of internal dials (**parameters**) so its guesses match real human text. At **inference** (when you chat), it turns your words into numbers, runs them through those dials, and produces a ranked list of likely next tokens — then picks one. No database of facts is consulted; it is pattern, not lookup.

ADVANTAGES

- + Fluent, flexible language on almost any topic
- + Adapts instantly to your instructions and tone
- + Reasons across ideas it was never explicitly taught

DISADVANTAGES

- Can state wrong things with total confidence (“hallucination”)
- No live knowledge of today unless connected to tools
- Same question can give slightly different answers

EXAMPLE

Type “The capital of France is” and the model rates “Paris” as by far the most likely next token, so that is what you get. Type “Write a poem about Paris” and thousands of tokens are plausible — which is why creative answers vary each time.



Fun Fact

The “GPT” in ChatGPT stands for Generative Pre-trained Transformer. The “Transformer” is a 2017 Google paper titled “Attention Is All You Need” — the whole AI boom traces back to it.

◆ FIND IT IN CLAUDE

+ menu → New chat

Watch next-token prediction live. Open a **New chat**, pick a model with the **model picker**, and type a half sentence. What you are watching is the prediction engine from this page, in real time.

THREE PROFESSIONAL PROMPTS, THREE LEVELS

LEVEL 1 · BEGINNER

FILL THE ORANGE

You are a patient teacher. Explain what a large language model is to someone completely new to AI in under 150 words. Use ONE everyday analogy (autocomplete or a weather forecast), no jargon, and end with one line on why it can be confidently wrong.

LEVEL 2 · INTERMEDIATE

FILL THE ORANGE

I have 5 minutes to explain LLMs to my team of non-engineers. Give me: a one-sentence definition, the 3 ideas they must leave with (token, prediction, training), a one-line analogy for each, and one myth to bust. One line per point.

LEVEL 3 · PROFESSIONAL

FILL THE ORANGE

Act as an AI educator designing a talk. Produce a 5-slide outline that moves a non-technical leadership audience from "AI is magic" to "AI predicts the next token". For each slide: title, the single idea, one concrete example, the misconception it corrects, and a 1-line speaker note. Output as a table.

BUILD IT UP — STEP BY STEP

- 1 Open claude.ai and start a new chat.
- 2 Type a half-finished sentence, e.g. "The best thing about mornings is", and send it.
- 3 Watch it complete your sentence — that is next-token prediction in front of you.
- 4 Now ask: "Explain what you just did, and why you chose those words."

EXECUTE IT

- 1 Ask the same factual question twice in two fresh chats; note if the wording differs.
- 2 Ask a question you know the answer to, and check it — feel how confident it sounds either way.
- 3 Rephrase one question three ways; see how the answer shifts with the phrasing.
- 4 Write one line in your notes: "AI predicts, it does not look up."

1.2

SESSION 1

THE LANGUAGE OF AI

Tokens, context windows, temperature, embeddings — the four words that unlock the rest.

**WHAT**

Four terms do most of the heavy lifting. A **token** is a chunk of text (roughly $\frac{3}{4}$ of a word). The **context window** is how much text the model can “hold in its head” at once. **Temperature** is a creativity dial — low is focused, high is wild. **Embeddings** turn words into numbers so the model can measure meaning.

WHY

These four explain the limits you keep bumping into: why very long documents get “forgotten” (context window), why answers vary (temperature), why the model groups related ideas (embeddings), and why you sometimes pay per token. Knowing them turns guesswork into control.

WHERE

Tokens show up on your bill and in “message too long” limits. The context window is why a long chat starts losing the thread. Temperature hides inside the API and some tools’ settings. Embeddings power search, memory, and “find similar” features.

WHEN

Mind the context window on long documents and long chats. Reach for temperature when output is too samey (raise it) or too random (lower it). You rarely set embeddings by hand — but you feel them every time search “just knows” what you meant.

HOW IT WORKS

Text is split into tokens, each mapped to an **embedding** — a long list of numbers that places its meaning in space, so “king” and “queen” sit close together. The model processes all tokens inside the context window together. When it picks the next token, **temperature** decides whether it always takes the top choice (low) or sometimes a surprising one (high).

ADVANTAGES

- + Context windows are huge now — whole books fit
- + Temperature gives you a real creativity control
- + Embeddings make semantic search feel like magic

DISADVANTAGES

- Long inputs cost more (more tokens = more money)
- Go past the window and early text is dropped
- High temperature can drift into nonsense

EXAMPLE

Paste a 40-page report and ask for a summary: it fits because modern context windows hold ~150,000+ tokens. Ask for “10 wildly different tagline ideas” and a higher temperature is what makes them genuinely different rather than ten near-copies.



Fun Fact

A rule of thumb: 1 token $\approx$ 4 characters $\approx$ $\frac{3}{4}$ of a word. So this Fun Fact box is about 40 tokens. Your name is probably one or two.

THREE PROFESSIONAL PROMPTS, THREE LEVELS

LEVEL 1 · BEGINNER

FILL THE ORANGE

Explain the difference between a "token" and a "context window" to a beginner using ONE notebook analogy. Under 120 words, no jargon, and finish with one practical tip for long chats.

LEVEL 2 · INTERMEDIATE

FILL THE ORANGE

I keep losing detail in long chats and long documents. In plain English, explain what a context window is, 3 signs I'm near the limit, and 3 habits that stop key facts being forgotten. Give the habits as a short checklist.

LEVEL 3 · PROFESSIONAL

FILL THE ORANGE

Act as an AI workflow engineer. For a document-summarising workflow, explain the token-cost vs quality trade-off, exactly when to raise or lower temperature, and a pre-flight checklist for keeping long inputs inside the context window without losing critical facts. Format: trade-off, temperature table, checklist.

BUILD IT UP — STEP BY STEP

- 1 In a chat, ask: "How many tokens is this sentence, roughly?" and paste any sentence.
- 2 Paste a long article and ask for a summary — notice it handles it in one go.
- 3 Ask the same creative question twice, once saying "be safe and predictable", once "be bold and varied".
- 4 Notice how the "bold" version feels like a higher temperature.

EXECUTE IT

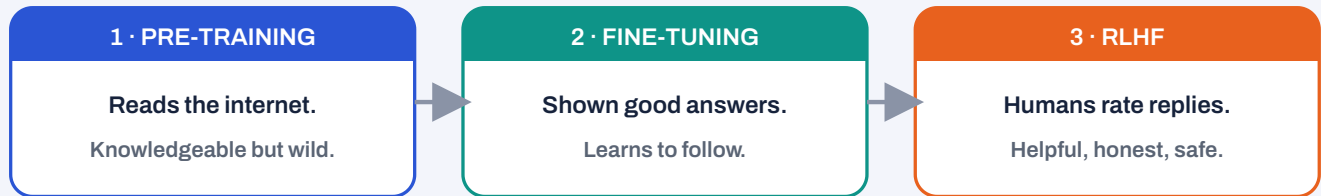
- 1 Before a long chat, paste the key facts once at the top so they stay in the window.
- 2 For long docs, ask for a summary first, then work from the summary to save tokens.
- 3 When answers feel repetitive, ask for "more variety"; when they wander, ask for "tighter, more focused".
- 4 Keep one reusable "context block" of your key facts to paste into new chats.

1.3

SESSION 1

HOW AI LEARNS

Pre-training, fine-tuning, and RLHF — the three stages that turn raw text into a helpful assistant.

**WHAT**

A model is built in three stages. **Pre-training**: it reads a vast amount of text and learns to predict tokens — raw, knowledgeable, but wild. **Fine-tuning**: it is shown examples of good answers so it learns to follow instructions. **RLHF** (Reinforcement Learning from Human Feedback): humans rate answers, and the model is nudged toward the helpful, honest, harmless ones.

WHY

This is **why Claude feels different from raw GPT**. The base skill is similar; the personality, safety, and helpfulness come from fine-tuning and RLHF. Understanding the stages tells you what a model can be great at (its training) and where it may be thin (what it never saw).

WHERE

You never run these stages — but you live with their results. Claude's careful, reasoning tone, its refusals, and its instruction-following are all products of Anthropic's fine-tuning and RLHF choices.

WHEN

Think about training whenever a model is oddly good or oddly bad at something. Great at code? It saw lots of code. Weak on last week's news? Training has a cut-off date — that is a training limit, not a bug.

HOW IT WORKS

Pre-training is expensive and done once, producing a “base model”. Fine-tuning shows it curated question-answer pairs. In RLHF, the model produces several answers, humans (or a trained reward model) rank them, and the model is updated to prefer the top-ranked style — repeated until it reliably behaves.

ADVANTAGES

- + Fine-tuning + RLHF make models genuinely helpful and safe
- + A model can be specialised for a domain
- + Alignment reduces harmful and nonsense outputs

DISADVANTAGES

- Knowledge is frozen at the training cut-off date
- Biases in the data can survive into the model
- RLHF can make a model overly cautious or agreeable

EXAMPLE

Ask a raw base model “How do I reset my password?” and it might continue with more *questions*, because it only learned to predict text. Ask Claude and you get clean steps — that helpful behaviour was taught in fine-tuning and RLHF.



Fun Fact

RLHF is why early assistants stopped replying to “Write me a poem” with another question. Humans quietly ranked millions of answers to teach the model good manners.

THREE PROFESSIONAL PROMPTS, THREE LEVELS

LEVEL 1 · BEGINNER

FILL THE ORANGE

Explain how an AI is "trained" to a beginner, using [an apprentice learning a trade] as the analogy. Give exactly 3 short steps (pre-training, fine-tuning, feedback) and one line on what each step adds.

LEVEL 2 · INTERMEDIATE

FILL THE ORANGE

Compare pre-training, fine-tuning and RLHF in a table with columns: what happens, who is involved, what it changes about the model, and one effect I'd actually notice. One line per cell.

LEVEL 3 · PROFESSIONAL

FILL THE ORANGE

Act as an ML educator advising a business. Explain why two models with similar pre-training can behave very differently (use fine-tuning and RLHF), then give 3 concrete implications for choosing a model for a sensitive customer-facing use case, and one question to ask any vendor.

BUILD IT UP — STEP BY STEP

- 1 Ask Claude: "What is your training cut-off date?" and note it.
- 2 Ask something from after that date; watch it tell you it may not know.
- 3 Now ask the same thing with web search / a connector on — see it fetch fresh info.
- 4 Ask: "Which of your behaviours come from RLHF?" for a plain explanation.

EXECUTE IT

- 1 For anything time-sensitive, assume the base model is out of date and give it fresh info or a tool.
- 2 For sensitive topics, expect careful, hedged answers — that is alignment working.
- 3 When a model refuses, rephrase to show intent; RLHF responds to clear, honest framing.
- 4 Note down each model's strengths as you find them — that is its training showing.

1.4

SESSION 1

READING THE SCOREBOARD

Benchmarks, the 3-way comparison, and why Claude leads for generalists.

MODEL	STRONG AT	FEEL	GENERALIST PICK
Claude	Reasoning, long docs, writing in your voice, safety	Thoughtful, follows instructions closely	Primary hub
ChatGPT (GPT-4o)	Speed, images, huge ecosystem, voice	Fast, chatty, eager	Everyday second
Gemini 1.5 Pro	Very long context, Google & image gen	Broad, Google-connected	Research & media

Rankings shift monthly — run your own 3-way test on your real task before you commit.

WHAT Models are ranked by **benchmarks** — standard tests. **MMLU** checks broad knowledge, **HumanEval** checks coding, and **arena ratings** let humans vote on which answer they prefer. No single number tells the whole story, so pros compare models on **their own** real task.

WHY Benchmark literacy stops you from believing hype. A model topping one test can be worse for *your* job. The skill that matters is running the same real prompt across Claude, ChatGPT and Gemini and judging the outputs yourself.

WHERE You will see benchmark claims in every launch announcement and comparison blog. The honest version lives in your own chat window, where you run the three side by side.

WHEN Run a comparison whenever the choice matters — a repeatable work task, a client deliverable, or before you commit to a paid plan. For a one-off question, just use the tool you have open.

HOW IT WORKS

Each benchmark feeds models the same questions and scores the answers. Arena ratings (like an Elo chess rating) pair two anonymous answers and ask people to pick the better one, aggregated over thousands of votes. For generalist professional work — reasoning depth, following instructions, long context, safety — Claude consistently ranks at or near the top.

ADVANTAGES

- + Benchmarks give a quick, comparable snapshot
- + Arena ratings capture real human preference
- + Your own test beats any leaderboard for your task

DISADVANTAGES

- Models can be “taught to the test”
- One score hides task-by-task differences
- Leaderboards shift monthly — today’s ranking ages fast

EXAMPLE

A model may win a coding benchmark yet write worse marketing copy than Claude. That is why the Week 1 build is a 3-way comparison: one identical prompt, three models, and your own verdict written into a Model Positioning Map.



Fun Fact

The famous “Chatbot Arena” ranks models like chess players with Elo scores — two anonymous answers appear, you vote for the better one, and millions of votes set the ladder.

THREE PROFESSIONAL PROMPTS, THREE LEVELS

LEVEL 1 · BEGINNER

FILL THE ORANGE

Give me ONE fair prompt I can paste into Claude, ChatGPT and Gemini to compare them on everyday writing, plus a 3-point checklist of what to look for in each answer. Keep it simple.

LEVEL 2 · INTERMEDIATE

FILL THE ORANGE

Design a fair 3-way model test for writing client emails.
Give me: the exact prompt to paste into all three, 4 scoring criteria (accuracy, tone, structure, usefulness) with a 1-5 rubric, and a blank scoring table I can fill in.

LEVEL 3 · PROFESSIONAL

FILL THE ORANGE

Act as an AI analyst. Build a reusable "Model Positioning Map" for my role: the 5 criteria that matter most, a 1-5 rubric for each, a scoring template for Claude / ChatGPT / Gemini on my real tasks, and a 3-line method to re-run it each quarter.
Output as a table plus instructions.

BUILD IT UP — STEP BY STEP

- 1 Pick one real task you do weekly and write it as a single clear prompt.
- 2 Open Claude, ChatGPT and Gemini in three tabs.
- 3 Paste the identical prompt into all three.
- 4 Put the three answers side by side in a document.

EXECUTE IT

- 1 Score each answer 1-5 on accuracy, tone, structure and usefulness.
- 2 Mark which model won for *this* task and why, in one line.
- 3 Repeat for a second task — winners often differ by task.
- 4 Save it as your Model Positioning Map; re-run it each quarter.

SESSION 1 DONE

HOW AI WORKS IS CLEAR.

You now have a working model of how AI models actually function — weights, prediction, training, and cost.

Next up: Session 2 — the Gen AI landscape, and picking the right tool for the job.

AI Curious → AI Aware

NEXT · SESSION 2

Session 1 was the engine. **Session 2** is the map — the six kinds of generative AI, how image models really work, and how to always find today's best tool. Continue in the Session 2 book.

menler

Your turning point in the AI era.